

Tanmay Mirgal

Full-Stack Engineer | ML/DL & MLOps Engineer | AI, Data Science & Computer Vision

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SUMMARY

Full-Stack and ML/DL Engineer with internship experience at Tata Advanced Systems Ltd. (TASL) and VigyBag, building production-ready MERN applications, ML pipelines, data science workflows, and computer vision systems. Multi-hackathon winner, published researcher, and active open-source contributor.

TECHNICAL SKILLS

Languages & Frameworks	JavaScript (ES6+), TypeScript, Python, R, React.js, Next.js, Node.js, Express.js, FastAPI
ML / AI / Data Science	TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost, OpenCV, RAG, Deep Learning, Predictive Modeling, Feature Engineering, EDA, Statistical Analysis, NLP, Time-Series Forecasting
MLOps & Pipelines	MLflow, Apache Airflow (Astronomer), DagsHub, GitHub Actions, Grafana
Cloud & DevOps	AWS (SageMaker, S3, RDS), Docker, Kubernetes, Nginx, CI/CD, Vercel, Linux
Databases	MongoDB, PostgreSQL, Redis, Convex, Supabase
Backend & Auth	REST APIs, WebSockets, WebRTC, JWT, OAuth 2.0, Prisma, Drizzle, Mongoose
Mobile & UI	React Native, Expo, Tailwind CSS, Shadcn/UI, Framer Motion, Redux
Tools	Postman, Jest, Jupyter Notebook, Jira, Bitbucket, Confluence, Git, Agile/Scrum

WORK EXPERIENCE

Tata Advanced Systems Ltd. (TASL)

Software Engineer Intern · Mumbai, India · 2024 – 2025

- Engineered CTS-71, a confidential high-fidelity naval simulation platform replicating real-world Combat Management Systems for the Indian Navy, built on the MERN stack with WebSockets.
- Authored Jest-based automated test suites for mission-critical modules; collaborated in Agile sprints using Jira, Bitbucket, and Confluence.

VigyBag

Open-Source Software Engineer – Top Contributor (2nd on Leaderboard) · Remote, India · 2024

- Built React.js UI components and Node.js backend features for an open-source e-commerce platform (182+ stars, 419+ forks) connecting rural artisans to urban consumers.
- Delivered product search, category filtering, and pricing comparison across 10+ categories; reviewed PRs and resolved issues following CI/CD guidelines.

PROJECTS

Slate – Next.js, TypeScript, React, SaaS

- Premium, enterprise-grade SaaS application designed to eliminate the friction between thinking, drawing, and documenting.

Live: <https://slate.tanmaymirgal.dev/> | GitHub: <https://github.com/Tanmay-Mirgal/Slate>

Orb – Next.js, Redis, PostgreSQL, MinIO, Docker, Better Auth, AWS (EC2)

- Self-hosted web application deployment platform — connects GitHub repositories, builds the app inside a sandboxed Docker container, and handles the rest end-to-end.
- Stores build artifacts in MinIO and routes live traffic through a built-in edge proxy with automatic SSL.

Live: <https://orb.tanmaymirgal.dev/> | GitHub: <https://github.com/Tanmay-Mirgal/Orb>

Samadhan (1st Place, Internal Hackathon 2025 – Vidyalankar Polytechnic) – Next.js, TypeScript, Expo, Convex, Tailwind CSS

- AI civic issue reporting ecosystem with real-time tracking and a cross-platform React Native app for citizen engagement.

Smart Meet – React, FastAPI, TensorFlow, OpenCV, Groq AI

- Hackathon-built meeting platform pairing a custom WebRTC engine with AI-driven transcription to turn live conversations into actionable summaries.
- Focused on low-friction setup and a high-speed operator experience for rapid deployment.

Live: <https://smart-meet.tanmaymirgal.dev/> | GitHub: <https://github.com/Tanmay-Mirgal/Smart-Meet>

Finch AI – *Next.js, Flask, XGBoost, PostgreSQL, MLflow, DagsHub*

- ML-driven banking recommendation engine for 500+ users; deployed XGBoost models with full MLOps pipelines (MLflow, DagsHub, AWS RDS) for continuous retraining and monitoring.
- Applied end-to-end data science workflow: EDA, feature engineering, model selection, hyperparameter tuning, and performance monitoring with Grafana dashboards.

Cognix – *React Native, Express.js, MongoDB, RAG, AWS SageMaker*

- Clinical cognitive testing app using RAG for intelligent question generation; deployed on AWS with SageMaker and automated retraining workflows orchestrated via Apache Airflow (Astronomer).

EDUCATION

Vidyalankar Polytechnic • Diploma in Computer Engineering · Mumbai, Maharashtra · 2023 – 2026

ACHIEVEMENTS & CERTIFICATIONS

- Winner – Industrial Hackathon 2026 (TATA): Built an enterprise-grade AI solution under competition conditions.
- Winner – Internal Hackathon 2025 (Vidyalankar Polytechnic): Built Samadhan, an AI civic reporting platform.
- Research Paper Published – JAAFR Journal (2025): Peer-reviewed AI and engineering research.
- INNOVGENIUS IDEATHON – Tata Consultancy Services (2025): National-level ideathon participant.
- Top Contributor – GSSOC 2024: Recognised among top contributors in GirlScript Summer of Code.
- Complete AI/ML & Data Science Bootcamp (Udemy)
- Generative AI for Software Development (IBM SkillsBuild)
- Intro to LLMs (Google Cloud)
- Postman API Fundamentals Student Expert